

300 Integral Practice Set

Maths Olympiad Preparation

2nd October 2024

§ Introduction

Welcome to our "*300 Integral Practice Set*," a carefully curated collection of problems designed to challenge and deepen your understanding of integration. Packed with 150 indefinite and 150 definite integration problems, this set, compiled over the course of 48 days, focuses entirely on providing a diverse range of problems without solutions, allowing you to sharpen your problem-solving skills and explore multiple approaches to each question.

This document is intended for learners who want to strengthen their grasp over integration & Calculus, in general, through hands-on practice. Whether you're preparing for competitive exams, academic tests, or simply aiming to master the subject, this set offers a comprehensive platform for self-assessment and skill enhancement. We encourage you to explore each problem independently, ponder upon various strategies, and enjoy the process of learning through discovery. Join us on Telegram *Maths Olympiad Preparation* and stay tuned for further updates and future releases!

Happy Problem Solving!

§ Indefinite Integrals

1.

$$\int \frac{x^3}{e^{2x^2}} dx = -\frac{2x^2 + 1}{8e^{2x^2}} + \mathcal{C}.$$

2.

$$\int \frac{\ln(1+x^2)}{x} dx = -\frac{\text{Li}_2(-x^2)}{2} + \mathcal{C}.$$

3.

$$\int \frac{1-2x}{\sqrt{x(1-x)(1-x+x^2)}} dx = 2 \arcsin \sqrt{x(1-x)} + \mathcal{C}.$$

4.

$$\int \frac{dx}{\sqrt[4]{\sin^3 x \cos^5 x}} = 4\sqrt[4]{\tan x} + \mathcal{C}.$$

5.

$$\int \frac{\sec x}{\tan x + \csc x} dx = \frac{1}{\sqrt{5}} \ln \left| \frac{2 \cos x - 1 - \sqrt{5}}{2 \cos x - 1 + \sqrt{5}} \right| + \mathcal{C}.$$

6.

$$\int \frac{\sin x - \cos x}{(\sin x + \cos x)\sqrt{\sin^2 x \cos^2 x + \sin x \cos x}} dx = -\cos^{-1} \left[\frac{1}{1 + \sin 2x} \right] + \mathcal{C}.$$

7.

$$\int \frac{1}{(1-x^2)\sqrt[4]{(2x^2-1)}} dx = \frac{1}{4} \ln \left| \frac{x + \sqrt[4]{(2x^2-1)}}{x - \sqrt[4]{(2x^2-1)}} \right| - \frac{1}{2} \arctan \left[\frac{\sqrt[4]{(2x^2-1)}}{x} \right] + \mathcal{C}.$$

8.

$$\int \frac{1}{1 + \sqrt{\tanh x}} dx = \frac{1}{2} \tan^{-1} \sqrt{\tanh x} - \frac{1}{4} \ln \left| \frac{\tanh x + 1}{\tanh x - 1} \right| + \frac{1}{2} \frac{1}{\sqrt{\tanh x + 1}} + \mathcal{C}.$$

9.

$$\int \frac{(2e^x + 1) \sin(x) - \cos(x) - 1}{e^{2x} + (\cos(x) - \sin(x) + 2)e^x + \cos(x) - \sin(x) \cos(x) - \sin(x) + 1} dx = \ln \left| \frac{e^x + 1 - \sin x}{e^x + 1 + \cos x} \right| + \mathcal{C}$$

10.

$$\int \frac{169 \sin x}{5 \sin x + 12 \cos x} dx = -12 \ln |5 \sin x + 12 \cos x| + 5x + \mathcal{C}.$$

11.

$$\int \frac{1}{\sqrt{\tanh(x) + 1}} dx = \frac{1}{\sqrt{2}} \left(\ln |e^x + \sqrt{e^{2x} + 1}| - \frac{\sqrt{e^{2x} + 1}}{e^x} \right) + \mathcal{C}.$$

12.

$$\int \frac{dx}{1 + \sin^6 x} = \frac{1}{3\sqrt{2}} \arctan(\sqrt{2} \tan x) + \frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3} \tan 2x}{2} \right) + \frac{1}{12} \ln \left| \frac{2 + \sin 2x}{2 - \sin 2x} \right| + \mathcal{C}.$$

13.

$$\int \frac{dx}{x^6 + 1} = \frac{1}{3} \tan^{-1} x + \frac{1}{6} \tan^{-1} \left(x - \frac{1}{x} \right) + \frac{1}{4\sqrt{3}} \ln \left| \frac{x^2 + \sqrt{3}x + 1}{x^2 - \sqrt{3}x + 1} \right| + \mathcal{C}.$$

14.

$$\int \frac{1}{\sqrt{1 + \sec^2(x)}} dx = \arcsin \left(\frac{\sin x}{\sqrt{2}} \right) + \mathcal{C}.$$

15.

$$\int \frac{1}{\sqrt{x^3 - x^4}} dx = -\cot \arcsin \sqrt{x} + \mathcal{C}.$$

16.

$$\int \sqrt{\coth x} dx = \frac{1}{2} \ln \left| \frac{\sqrt{\coth x} + 1}{\sqrt{\coth x} - 1} \right| - \tan^{-1} \sqrt{\coth x} + \mathcal{C}.$$

17.

$$\int \frac{1}{(x^2 + 4)^3} dx = \frac{3x}{128(x^2 + 4)} + \frac{x}{16(x^2 + 4)^2} + \frac{3 \arctan(\frac{x}{2})}{256} + \mathcal{C}.$$

18.

$$\int \tan^{-1}(x^2) dx = x \tan^{-1}(x^2) - \frac{\sqrt{2}}{4} \ln \left| \frac{x^2 - x\sqrt{2} + 1}{x^2 + x\sqrt{2} + 1} \right| - \frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2 - 1}{x\sqrt{2}} \right) + \mathcal{C}.$$

19.

$$\int \frac{e^{2x} - e^x + 1}{(e^x \sin x + \cos x)(e^x \cos x - \sin x)} dx = \ln \left| \frac{e^x \sin x + \cos x}{e^x \cos x - \sin x} \right| + \mathcal{C}.$$

20.

$$\int \frac{\sin^3 x}{\cos^8 x} dx = \frac{1}{7 \cos^7 x} - \frac{1}{5 \cos^5 x} + \mathcal{C}.$$

21.

$$\int \frac{2x^3 - 1}{x(x^3 + 1)} dx = \ln \left| \frac{x^3 + 1}{x} \right| + \mathcal{C}.$$

22.

$$\int \frac{\cos 5x - \cos 4x}{1 - \cos 3x} dx = 2 \sin x - \sin 2x - \sqrt{3} \ln \left| \frac{\sqrt{3} + \tan(\frac{x}{2})}{\sqrt{3} - \tan(\frac{x}{2})} \right| + \mathcal{C}.$$

23.

$$\int \frac{x^2 - 1}{(x^2 + 1)\sqrt{x^4 + 1}} dx = \frac{1}{\sqrt{2}} \arctan \frac{\sqrt{x^4 + 1}}{x\sqrt{2}} + \mathcal{C}.$$

24.

$$\int 6x^2 \sin(x^3 + 1) dx = -2 \cos(x^3 + 1) + \mathcal{C}.$$

25.

$$\int \sin(4x) e^{\tan^2 x} dx = -2 \cos^4 x e^{\tan^2 x} + \mathcal{C}.$$

26.

$$\int \frac{1}{(x^4 - 1)^2} dx = \frac{3}{16} \ln \left| \frac{1+x}{1-x} \right| + \frac{3}{8} \tan^{-1} x - \frac{x}{4(x^4 - 1)} + \mathcal{C}.$$

27.

$$\int \frac{1}{(1 + \cosh ax)^2} dx = \frac{4}{a} \cdot \frac{-3e^{ax} - 1}{3(e^{ax} + 1)^3} + \mathcal{C}.$$

28.

$$\int \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{x}{2} + \frac{1}{2} \ln |\cos x| + \ln |\sqrt{\tan x} + 1| + \frac{\sqrt{2}}{4} \ln \left| \frac{\tan x - \sqrt{2 \tan x} + 1}{\tan x + \sqrt{2 \tan x} + 1} \right| + \mathcal{C}.$$

29.

$$\int \frac{\sqrt{1+x^4}}{1-x^4} dx = \frac{\sqrt{2}}{4} \left[\arctan \frac{x\sqrt{2}}{\sqrt{1+x^4}} - \frac{1}{2} \ln \left| \frac{x\sqrt{2} - \sqrt{1+x^4}}{x\sqrt{2} + \sqrt{1+x^4}} \right| \right] + \mathcal{C}.$$

30.
$$\int \frac{\cos 2x \sin 4x}{\cos^4 x (1 + \cos^2 2x)} dx = 2 \ln |1 + \cos 2x| + \frac{2}{1 + \cos 2x} - \ln |1 + \cos^2 2x| + \mathcal{C}.$$

31.
$$\int \frac{dx}{\sqrt[3]{x^2(1 - \sqrt[3]{x})^4}} = \frac{9}{2} \sqrt[3]{\frac{\sqrt[3]{x} - 1}{\sqrt[3]{x} + 1}} + \mathcal{C}.$$

32.
$$\int \frac{e^x(2e^x + 1)}{(e^{2x} + e^x + 1)\sqrt{e^{2x} + e^x}} dx = \arctan \sqrt{e^{2x} + e^x} + \mathcal{C}.$$

33.
$$\int \frac{1 - x - x^2}{e^x(4 - 4x + x^2)} dx = \frac{x - 1}{e^x(x - 2)} + \mathcal{C}.$$

34.
$$\int e^x(3 \sin x - \cos x) dx = \frac{2\sqrt{e^x}}{\sin x + \cos x} + \mathcal{C}.$$

35.
$$\int \frac{e^x(x^6 - 4x^5 + x^4 - 2x^3 + x + 1)}{\sqrt{(x^5 + x^3 + 1)^3}} dx = \frac{xe^x}{\sqrt{x^5 + x^3 + 1}} + \mathcal{C}.$$

36.
$$\int \frac{xe^x}{x + 1} dx = e^x - \frac{\text{Ei}(x + 1)}{e} + \mathcal{C}.$$

37.
$$\int \frac{\sin(x)}{\cos^3(x) + \sin(x) \cos^2(x)} dx = \tan x - \ln |1 + \tan x| + \mathcal{C}.$$

38.
$$\int \frac{e^x(2 - 2x^2 - x)}{2x^2\sqrt{x + 1}} dx = -\frac{e^x\sqrt{x + 1}}{x} + \mathcal{C}.$$

39.
$$\int x\sqrt{x + 2} dx = \frac{2}{15}(3x - 4)\sqrt{(x + 2)^3} + \mathcal{C}.$$

40.
$$\int \tan(x) \ln(\cos(x)) dx = -\frac{(\ln |\cos x|)^2}{2} + \mathcal{C}.$$

41.
$$\int \tan^4(x) \sec^6(x) dx = \frac{\tan^9 x}{9} + \frac{2 \tan^7 x}{7} + \frac{\tan^5 x}{5} + \mathcal{C}.$$

42.
$$\int \frac{x^4 - 1}{x^2\sqrt{x^4 - x^2 + 1}} dx = \frac{\sqrt{x^4 - x^2 + 1}}{x} + \mathcal{C}.$$

43.
$$\int \frac{e^x(2 - x^2)}{(1 - x)\sqrt{1 - x^2}} dx = e^x \cdot \frac{\sqrt{1 + x}}{\sqrt{1 - x}} + \mathcal{C}.$$

44.
$$\int (\sin(\arctan x) \cdot \ln x) dx = \sqrt{1 + x^2} \ln x + \tanh^{-1}(\sqrt{1 + x^2}) - \sqrt{1 + x^2} + \mathcal{C}.$$

45.
$$\int e^{\sqrt{\sin x}} \sqrt{\sin 2x} dx = 2\sqrt{2}e^{\sqrt{\sin x}}(\sin x - 2\sqrt{\sin x} + 2) + \mathcal{C}.$$

46.

$$\int \frac{\sin(n+1)x \sin nx}{\sin x} dx = -\sum_{k=1}^n \frac{\cos 2kx}{2k} + \mathcal{C}.$$

47.

$$\int \ln(\sqrt{1-x} + \sqrt{x+1}) dx = \frac{1}{2} \left(x(\ln 2 - 1) + x \ln(1 + \sqrt{1-x^2}) + \sin^{-1} x \right) + \mathcal{C}.$$

48.

$$\int \frac{\sin^3 x dx}{(\cos^4 x + 3 \cos^2 x + 1) \tan^{-1}(\sec x + \cos x)} = \ln |\tan^{-1}(\sec x + \cos x)| + \mathcal{C}.$$

49.

$$\int x^2 \sqrt{1+x^2} dx = \frac{1}{32} \sinh(4 \sinh^{-1} x) - \frac{1}{8} \sinh^{-1} x + \mathcal{C}.$$

50.

$$\int \frac{1}{\cos x \sqrt{\cos 2x}} dx = \arctan \frac{\sin x}{\sqrt{\cos 2x}} + \mathcal{C}.$$

51.

$$\int \cos(\ln x) dx = \frac{1}{2} (x \cos(\ln x) + x \sin(\ln x)) + \mathcal{C}.$$

52.

$$\int \frac{e^{2x}}{\sinh^4 x} dx = -\frac{8}{3} \cdot \frac{1 - 3e^{2x} + 3e^{4x}}{(e^{2x} - 1)^3} + \mathcal{C}.$$

53.

$$\int \sqrt[3]{1 + \sqrt[3]{x}} dx = 9(1 + \sqrt[3]{x}) \sqrt[3]{1 + \sqrt[3]{x}} \left(\frac{1}{10}(1 + 2\sqrt[3]{x} + 2\sqrt[3]{x}) - \frac{2}{7}(1 + \sqrt[3]{x}) + \frac{1}{4} \right) + \mathcal{C}.$$

54.

$$\int \frac{dx}{\sin x + \sec x} = \tan^{-1} \left[\frac{2 + 2 \tan \frac{x}{2}}{1 + 2 \tan \frac{x}{2} - 2 \tan^2 \frac{x}{2}} \right] + \ln \left| \frac{\cos x - \sin x - \sqrt{3}}{\cos x - \sin x + \sqrt{3}} \right| + \mathcal{C}.$$

55.

$$\int \frac{e^{-ax}}{x^2} dx = -\frac{ax \operatorname{Ei}(-ax) + e^{-ax}}{x} + \mathcal{C}.$$

56.

$$2 \int \frac{x \ln x}{1+x^2} dx = \operatorname{Li}_2(ix) + \operatorname{Li}_2(-ix) + \ln |x| \ln |1+x^2| + \mathcal{C}.$$

57.

$$4 \int \frac{\sqrt{x^4 + x^2 + 1}}{x} dx = \sinh^{-1} \left(\frac{2x^2 + 1}{\sqrt{3}} + 2\sqrt{x^4 + x^2 + 1} - 2 \tanh^{-1} \frac{x^2 + 2}{2\sqrt{x^4 + x^2 + 1}} \right) + \mathcal{C}.$$

58.

$$\int x \ln(1+x^3) dx = \frac{1}{2} x \ln(1+x^3) - \frac{3}{4} x^2 + \frac{1}{4} \ln(x^4 - x^2 + 1) - \frac{1}{2} \ln(1+x) + \frac{\sqrt{3}}{2} \arctan \left(\frac{2x+1}{\sqrt{3}} \right) + \mathcal{C}.$$

59.

$$\int \frac{x \ln x}{\sqrt{(x^2-1)^3}} = \arctan \sqrt{x^2-1} - \frac{\ln x}{\sqrt{x^2-1}} + \mathcal{C}.$$

60.

$$\int \frac{(x^2-1)dx}{x\sqrt{x^4+3x^2+1}} = \ln \left| \frac{x^2+1+\sqrt{x^4+3x^2+1}}{x} \right| + \mathcal{C}.$$

61.

$$\int \frac{dx}{(2x-3)\sqrt{4x-x^2}} = -\frac{1}{\sqrt{15}} \ln \left| \frac{x+6+\sqrt{60x-15x^2}}{2x+3} \right| + C.$$

62.

$$\int \frac{xe^x dx}{\sqrt{1+e^x}} = (2x-4)\sqrt{1+e^x} - 2 \ln \left| \frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1} \right| + C.$$

63.

$$\frac{\arctan x}{x^4} dx = \frac{1}{6} \ln \left| \frac{1+x^2}{x^2} \right| - \frac{\arctan x}{3x^3} - \frac{1}{6x^2} + C.$$

64.

$$\int \frac{\arctan x}{(1+x)^3} dx = \frac{1}{4} \frac{\ln|x+1|}{\sqrt{x^2+1}} - \frac{\arctan x}{2(x+1)^2} - \frac{1}{4(x+1)} + C.$$

65.

$$\int \frac{dx}{\sqrt{1+e^x+e^{2x}}} = \ln \left| \frac{1+e^x-\sqrt{1+e^x+e^{2x}}}{1+e^x+\sqrt{1+e^x+e^{2x}}} \right| + C.$$

66.

$$\int \frac{(3+x^2)^2 x^3 dx}{(1+x^2)^3} = \frac{1}{2} x^2 + \frac{3}{2} \ln(1+x^2) + \frac{1}{(1+x^2)^2} + C.$$

67.

$$\int \frac{x^2-8x+7}{(x^2-3x-10)^2} dx = \frac{8}{49(x-5)} - \frac{27}{49(x+2)} + \frac{30}{343} \ln \left| \frac{x-5}{x+2} \right| + C.$$

68.

$$\int \frac{(x^2-1)dx}{(x^2+1)\sqrt{1+x^4}} = \frac{1}{\sqrt{2}} \arccos \frac{x\sqrt{2}}{x^2+1} + C.$$

69.

$$\int \frac{x^2(x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx = 2 \ln |x \sin x + \cos x| - \frac{x^2}{x \tan x + 1} + C.$$

70.

$$\int x \sqrt[5]{1+x} dx = \frac{5}{66} (6x-5) \sqrt[5]{(1+x)^6} + C.$$

71.

$$\int (1+x^2) \sqrt[3]{1+x} = \frac{3}{70} (7x^2-6x+22) \sqrt[3]{(1+x)^4} + C.$$

72.

$$6 \int \sqrt{2x+1} \arctan(\sqrt{2x+1}) dx = \ln|x+1| + 2 \arctan(\sqrt{2x+1}) \sqrt{(2x+1)^3} - 2x + C.$$

73.

$$8 \int x \arctan(2x^2+1) dx = (4x+2) \arctan(2x+1) - \ln|2x^4+2x^2+1| + C.$$

74.

$$\int \arctan[(x^2+1)(x^2+2)] dx = x \arctan(x^4+3x^2+2) - \frac{1}{2} \sum_{\omega: \omega^8+6\omega^6+13\omega^4+12\omega^2+5=0} \frac{\ln|x-\omega|}{\omega^4+3\omega^2+2} + C.$$

75.

$$\int \frac{\sqrt{e^x} \cos x dx}{\sqrt[3]{3 \cos x + 4 \sin x}} = \frac{6\sqrt{e^x}}{25} \sqrt[3]{(3 \cos x + 4 \sin x)^2} + C.$$

76.

$$\int x \tanh^2 x dx = \frac{x^2}{2} - x \tanh x + \ln |\cosh x| + \mathcal{C}.$$

77.

$$\int \frac{x^3}{1+x^6} dx = \frac{1}{12} \ln \left| \frac{x^4 - x^2 + 1}{x^4 + 2x^2 + 1} \right| + \frac{\sqrt{3}}{6} \arctan \left(\frac{\sqrt{3}}{1+2x^2} \right) + \mathcal{C}.$$

78.

$$\int \frac{dx}{\cos x + \csc x} = \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - \sin x + \cos x} \right| - \arctan(\sin x + \cos x) + \mathcal{C}.$$

79.

$$\int \frac{\tan(45^\circ - x)}{\cos^2 x \sqrt{\tan^3 x + \tan^2 x + \tan x}} dx = -2 \arctan(\sqrt{\tan x + 1 + \cot x}) + \mathcal{C}.$$

80.

$$\int \frac{x \ln x}{1 - \ln x} dx = -e^2 \operatorname{Ei}(2 \ln x - 2) - \frac{x^2}{2} + \mathcal{C}.$$

81.

$$4 \int \sqrt{(x^2 + a^2)^3} dx = x \sqrt{(x^2 + a^2)^3} + \frac{3a^2 x}{2} \sqrt{x^2 + a^2} + \frac{3a^4}{2} \ln |x + \sqrt{x^2 + a^2}| + \mathcal{C}.$$

82.

$$\int \frac{1}{x} \sqrt{\frac{1-x}{1+x}} dx = \ln \left| \frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right| + 2 \arctan \sqrt{\frac{1-x}{1+x}} + \mathcal{C}.$$

83.

$$\int (\arcsin x)^2 dx = x(\arcsin x)^2 + 2 \arcsin x \sqrt{1-x^2} - 2x + \mathcal{C}.$$

84.

$$\int \frac{dx}{ae^{mx} + be^{-mx}} = \frac{-1}{m\sqrt{ab}} \arctan \left(e^{mx} \sqrt{\frac{a}{b}} \right) + \mathcal{C}.$$

85.

$$\int \frac{dx}{1+x^3} = \frac{1}{3} \ln \left(\frac{x+1}{\sqrt{x^2-x+1}} \right) + \frac{1}{\sqrt{3}} \arctan \left(\frac{2x-1}{\sqrt{3}} \right) + \mathcal{C}.$$

86.

$$24 \int \frac{x^3 dx}{(1+8x^3)^2} = \frac{1}{12} \ln \left| \frac{(1+2x)^2}{1-2x+4x^2} \right| + \frac{\sqrt{3}}{6} \arctan \left(\frac{4x-1}{\sqrt{3}} \right) - \frac{x}{1+8x^3} + \mathcal{C}.$$

87.

$$\int \frac{dx}{x^4 + x^2 + 1} = \frac{1}{4} \ln \left| \frac{x^2 + x + 1}{x^2 - x + 1} \right| + \frac{1}{2\sqrt{3}} \arctan \left(\frac{\sqrt{3}x}{1-x^2} \right) + \mathcal{C}.$$

88.

$$\int \frac{(2ax+b)e^{\sqrt{\ln(ax^2+bx+c)}} dx}{(ax^2+bx+c)\sqrt{\ln(ax^2+bx+c)}} = 2e^{\sqrt{\ln(ax^2+bx+c)}} + \mathcal{C}.$$

89.

$$\int \frac{\sqrt{ax^2+bx+c}}{x} dx = 2\sqrt{ax^2+bx+c} \left(1+b+2\sqrt{a}x-2\sqrt{ax^2+bx+c} \right) + \mathcal{C}.$$

90.

$$\int \frac{x^7+2}{(x^2+x+1)^2} = \frac{x}{x^2+x+1} + \frac{2}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} - 2 \ln |x^2+x+1| + \frac{x^4}{4} - \frac{2x^3}{3} + \frac{x^2}{2} + 2x + \mathcal{C}.$$

91.

$$\int \frac{dx}{(x^2 + 2x + 10)^3} = \frac{1}{648} \left(\arctan \frac{x+1}{3} + \frac{3(x+1)}{x^2 + 2x + 10} + \frac{18(x+1)}{(x^2 + 2x + 10)^2} \right) + \mathcal{C}.$$

92.

$$\int \frac{x^5 - x^4 - 26x^2 - 24x - 25}{(x^2 + 4x + 5)^2(x^2 + 4)^2} dx = -\frac{x}{8(x^2 + 4)} - \frac{2x + 5}{2(x^2 + 4x + 5)} - \frac{1}{16} \arctan \frac{x}{2} - \arctan(x + 2) + \mathcal{C}.$$

93.

$$\int \frac{\sqrt{1-x^3}}{x^2\sqrt{x}} dx = -\frac{2}{3} \sqrt{\frac{1-x^2}{x^3}} - \frac{2}{3} \arcsin \sqrt{x^3} + \mathcal{C}.$$

94.

$$\int \frac{\ln x - \ln(x+1)}{x(x+1)} dx = \frac{1}{2} \ln^2 \left(1 + \frac{1}{x} \right) + \mathcal{C}.$$

95.

$$\int \arccos \sqrt{\frac{x}{x+1}} dx = x \arccos \sqrt{\frac{x}{x+1}} + \sqrt{x} - \arctan \sqrt{x} + \mathcal{C}.$$

96.

$$\int \frac{dx}{\sqrt[4]{1+x^4}} = \frac{1}{4} \ln \left| \frac{\sqrt[4]{1+x^4} + x}{\sqrt[4]{1+x^4} - x} \right| - \frac{1}{2} \arctan \frac{\sqrt[4]{1+x^4}}{x} + \mathcal{C}.$$

97.

$$\int \frac{(x+1)^4 dx}{(x^2 + 2x + 2)^3} = \frac{3}{8} \arctan(x+1) - \frac{5x^3 + 5x^2 + 18x + 18}{8(x^2 + 2x + 2)^2} + \mathcal{C}.$$

98.

$$\int \frac{\sqrt[3]{1+x^3}}{x^2} dx = \frac{1}{\sqrt{3}} \arctan \frac{2\sqrt[3]{1+x^3} + x}{\sqrt{3}x} - \frac{\sqrt[3]{1+x^3}}{x} - \frac{1}{3} \ln \left| \frac{\sqrt[3]{1+x^3} + x}{\sqrt{\sqrt[3]{(1+x^3)^2} + x\sqrt[3]{1+x^3} + x^2}} \right| + \mathcal{C}.$$

99.

$$\int \frac{dx}{x^{11}\sqrt{1+x^4}} = -\frac{1}{10} \sqrt{\left(\frac{1+x^4}{x^4}\right)^5} + \frac{1}{3} \sqrt{\left(\frac{1+x^4}{x^4}\right)^3} - \frac{1}{2} \sqrt{\frac{1+x^4}{x^4}} + \mathcal{C}.$$

100.

$$\int \frac{dx}{\sqrt{(1-x^2)(3-2x-x^2)}} = \frac{1}{2\sqrt{2}} \tanh^{-1} \left(\frac{3x+5}{2\sqrt{2}\sqrt{x^2+4x+3}} \right) + \mathcal{C}.$$

101.

$$\int \cosh^2(3x) \tanh(2x) dx = \frac{1}{4} \ln |\cosh 2x| + \frac{1}{12} \cosh(6x) - \frac{1}{2} \cosh(2x) + \mathcal{C}.$$

102.

$$\int \frac{dx}{\sqrt{\sin x(1-\sin x)}} = \sqrt{2} \ln \left| \frac{\sqrt{\sec x - 1} - \sqrt{\tan x}}{\sqrt{\sec x - 1} + \sqrt{\tan x}} \right| + \mathcal{C}.$$

103.

$$\int \cot^{-1}(x^2 + x + 1) dx = x \cot^{-1}(x^2 + x + 1) + \frac{1}{2} \ln \frac{x^2 + 1}{x^2 + 2x + 2} + \tan^{-1}(x + 1) + \mathcal{C}.$$

104.

$$\int \frac{(x+1)^3(x-1)}{(x^2+1)^2\sqrt{x^4+x^2+1}} dx = \tan^{-1} \left(\sqrt{\left(x + \frac{1}{x}\right)^2 - 1} \right) + \frac{2\sqrt{\left(x + \frac{1}{x}\right)^2 - 1}}{\left(x + \frac{1}{x}\right)} + \mathcal{C}.$$

105.

$$\int \csc^2 x \ln \left(\cos x + \sqrt{\cos 2x} \right) dx = -\cot x \cdot \ln \left| \cos x + \sqrt{\cos(2x)} \right| + \frac{\sqrt{\cos(2x)}}{\sin x} - \cot x - x + \mathcal{C}.$$

106.

$$\int \frac{\sin^3 \frac{x}{2}}{\cos \frac{x}{2} \sqrt{\cos x + \cos^2 x + \cos^3 x}} dx = \tan^{-1} \sqrt{\cos x + \sec x + 1} + \mathcal{C}.$$

107.

$$\int e^{13x} \sin 2x \cos 5x dx = \frac{e^{13x}}{38804} (1157 \sin(7x) - 623 \cos(7x) - 1147 \sin(3x) + 327 \cos(3x)) + \mathcal{C}.$$

108.

$$\int x \sqrt{\frac{a^2 - x^2}{a^2 + x^2}} dx = \frac{1}{2} \sqrt{a^4 - x^4} - a^2 \csc^{-1} \left(\frac{\sqrt{2}a}{\sqrt{a^2 - x^2}} \right) + \mathcal{C}.$$

109.

$$\int x \csc x dx = i [\operatorname{Li}_2(-e^{ix}) - \operatorname{Li}_2(e^{ix})] + x \ln \left| \frac{1 - e^{ix}}{1 + e^{ix}} \right| + \mathcal{C}.$$

110.

$$96 \int \frac{x^7 dx}{\sqrt{x^4 + x^2 + 1}} = 2(8x^4 - 10x^2 + 1) \sqrt{x^4 + x^2 + 1} + 21 \sinh^{-1} \left(\frac{2x^2 + 1}{\sqrt{3}} \right) + \mathcal{C}.$$

111.

$$\int e^{x \sin x + \cos x} \cdot \left(\frac{x^4 \cos^3 x - x \sin x + \cos x}{x^2 \cos^2 x} \right) dx = e^{x \sin x + \cos x} \left(x - \frac{1}{x \cos x} \right) + \mathcal{C}$$

112.

$$\int x^6 e^{2x} dx = \frac{x^6 e^{2x}}{2} - \frac{3x^5 e^{2x}}{2} + \frac{15x^4 e^{2x}}{4} - \frac{15x^3 e^{2x}}{2} + \frac{45x^2 e^{2x}}{4} - \frac{45x e^{2x}}{4} + \frac{45e^{2x}}{8} + \mathcal{C}$$

113.

$$\int \frac{\sin^2 x}{\cos^2 x + \cot^2 x} dx = \frac{1}{2} \tan x - x + \frac{\sqrt{2}}{4} \arctan(\sqrt{2} \tan x) + \mathcal{C}$$

114.

$$\int x^2 e^{\sin^{-1} x} dx = e^{\arcsin x} \left(\frac{3}{10} x^3 - \frac{1}{10} x + \frac{1}{5} \sqrt{1 - x^2} - \frac{1}{10} (1 - x^2)^{\frac{3}{2}} \right) + \mathcal{C}.$$

115.

$$\int \left[\sqrt[6]{\frac{x}{x-2}} - \sqrt[4]{\frac{x}{x-2}} \right] \cdot \frac{dx}{x^2 - 2x} = 2 \sqrt[4]{\frac{x}{x-2}} - 3 \sqrt[6]{\frac{x}{x-2}} + \mathcal{C}.$$

116.

$$\int \left(\operatorname{erf}(x) \right)^2 dx = x \left(\operatorname{erf}(x) \right)^2 + \frac{2}{\sqrt{\pi}} \operatorname{erf}(x) e^{-x^2} - \sqrt{\frac{2}{\pi}} \operatorname{erf}(\sqrt{2}x) + \mathcal{C}.$$

117.

$$\int \frac{dx}{\arcsin^3 x} = \frac{x \arcsin x - \sqrt{1 - x^2} - \arcsin^2 x \operatorname{Ci}(\arcsin x)}{2 \arcsin^2 x} + \mathcal{C}.$$

118.

$$\int \mathcal{L}_t(\tan t)(s) dt = \frac{x \operatorname{Ci}(s) \sin s}{s} + \frac{x(\pi - 2 \operatorname{Si}(s)) \cos s}{2s} + \mathcal{C}.$$

119.

$$\begin{aligned} \int \frac{dx}{x^{2^n} + 1} &= \frac{1}{2^{n-1}} \sum_{k=1}^{2^{n-1}} \sin \left(\frac{2k-1}{2^n} \pi \right) \arctan \left[\left(x - \cos \left(\frac{2k-1}{2^n} \pi \right) \right) \csc \left(\frac{2k-1}{2^n} \pi \right) \right] \\ &\quad - \frac{1}{2} \cos \left(\frac{2k-1}{2^n} \pi \right) \ln \left| x^2 - 2x \cos \left(\frac{2k-1}{2^n} \pi \right) + 1 \right| + \mathcal{C}. \end{aligned}$$

120.
$$\int \frac{x^2}{(ax+b)^n} dx = \frac{1}{a^3} \left(-\frac{(ax+b)^{3-n}}{(n-3)} + \frac{2b(ax+b)^{2-n}}{(n-2)} - \frac{b^2(ax+b)^{1-n}}{(n-1)} \right) + \mathcal{C} \quad (\text{for } n \notin \{1, 2, 3\}).$$

121.
$$\int \frac{1}{x^2(ax+b)^2} dx = -a \left(\frac{1}{b^2(ax+b)} + \frac{1}{ab^2x} - \frac{2}{b^3} \ln \left| \frac{ax+b}{x} \right| \right) + \mathcal{C}.$$

122. Prove that for $r = \sqrt{a^2 + x^2}$,

$$\int x^4 r^3 dx = \frac{x^3 r^5}{8} - \frac{a^2 x r^5}{16} + \frac{a^4 x r^3}{64} + \frac{3a^6 x r}{128} + \frac{3a^8}{128} \ln(x+r) + \mathcal{C}.$$

123. Prove that for $n > m \geq 0$ and $s = \sqrt{x^2 - a^2}$,

$$\int \frac{x^{2m} dx}{s^{2n+1}} = (-1)^{n-m} \frac{1}{a^{2(n-m)}} \sum_{i=0}^{n-m-1} \frac{1}{2(m+i)+1} \binom{n-m-1}{i} \frac{x^{2(m+i)+1}}{s^{2(m+i)+1}} + \mathcal{C}.$$

124.
$$\int (\sin b_1 x)(\sin b_2 x) dx = \frac{\sin((b_2 - b_1)x)}{2(b_2 - b_1)} - \frac{\sin((b_1 + b_2)x)}{2(b_1 + b_2)} + \mathcal{C} \quad (\text{for } |b_1| \neq |b_2|)$$

125.
$$12 \int \frac{x^3 dx}{1-x^6} = \ln \left| \frac{x^4 + x^2 + 1}{x^4 - 2x^2 + 1} \right| + 2\sqrt{3} \arctan \left(\frac{\sqrt{3}}{2x^2 + 1} \right) + \mathcal{C}.$$

126.
$$\int x^m \operatorname{arccot}(ax) dx = \frac{x^{m+1} \operatorname{arccot}(ax)}{m+1} + \frac{a}{m+1} \int \frac{x^{m+1}}{a^2 x^2 + 1} dx + \mathcal{C}, \quad (m \neq -1).$$

127.
$$\int (\sinh ax)(\sinh bx) dx = \frac{1}{a^2 - b^2} (a(\sinh bx)(\cosh ax) - b(\cosh bx)(\sinh ax)) + \mathcal{C}, \quad (\text{for } a^2 \neq b^2).$$

128.
$$\int \sinh(ax+b) \sin(cx+d) dx = \frac{a \cosh(ax+b) \sin(cx+d)}{a^2 + c^2} - \frac{c \sinh(ax+b) \cos(cx+d)}{a^2 + c^2} + \mathcal{C}.$$

129.
$$\int \sinh(ax+b) \cos(cx+d) dx = \frac{a \cosh(ax+b) \cos(cx+d)}{a^2 + c^2} + \frac{c \sinh(ax+b) \sin(cx+d)}{a^2 + c^2} + \mathcal{C}.$$

130.
$$\int \operatorname{arsinh}(ax)^2 dx = 2x + x \operatorname{arsinh}(ax)^2 - \frac{2\sqrt{a^2 x^2 + 1} \operatorname{arsinh}(ax)}{a} + \mathcal{C}.$$

131.
$$\int \cosh(ax+b) \sin(cx+d) dx = \frac{a \sinh(ax+b) \sin(cx+d)}{a^2 + c^2} - \frac{c \cosh(ax+b) \cos(cx+d)}{a^2 + c^2} + \mathcal{C}.$$

132.
$$\int \cosh(ax+b) \cos(cx+d) dx = \frac{a \sinh(ax+b) \cos(cx+d)}{a^2 + c^2} + \frac{c \cosh(ax+b) \sin(cx+d)}{a^2 + c^2} + \mathcal{C}.$$

133.
$$\int e^{cx} \varphi(bx)^n dx = \frac{e^{\frac{c^2}{2nb^2}}}{b\sqrt{n}(2\pi)^{n-1}} \Phi \left(\frac{b^2 xn - c}{b\sqrt{n}} \right) + \mathcal{C}, \quad b \neq 0, n > 0$$

134.
$$\int e^{ax} \left((a)^n f(x) - (-1)^n \frac{d^n f(x)}{dx^n} \right) dx = e^{ax} \sum_{k=1}^n (a)^{n-k} (-1)^{k-1} \frac{d^{k-1} f(x)}{dx^{k-1}} + \mathcal{C}.$$

135.

$$\int \frac{ae^{cx} - 1}{be^{cx} - 1} dx = \frac{(a-b)\ln(1-be^{cx})}{bc} + x + \mathcal{C}.$$

136.

$$\int x\Phi(a+bx) dx = \frac{1}{2b^2} ((b^2x^2 - a^2 - 1)\Phi(a+bx) + (bx-a)\varphi(a+bx)) + \mathcal{C}.$$

137.

$$\int \varphi(a+bx)^n dx = \frac{1}{b\sqrt{n(2\pi)^{n-1}}} \Phi(\sqrt{n}(a+bx)) + \mathcal{C}.$$

138.

$$\int x^2\varphi(a+bx) dx = \frac{1}{b^3} ((a^2+1)\Phi(a+bx) + (a-bx)\varphi(a+bx)) + \mathcal{C}.$$

139.

$$\int \frac{\ln(1+x)}{x(x-1)} dx = \text{Li}_2(-x) + \ln 2 \ln(x-1) - \text{Li}_2\left(\frac{1-x}{2}\right) + \mathcal{C}.$$

140.

$$2 \int \frac{\sin x dx}{\sec^4 x - 1} = 2 \cos x + \ln \left| \tan \frac{x}{2} \right| - \arctan(\cos x) + \mathcal{C}.$$

141.

$$\int \frac{x^2 dx}{(x-a)(x-b)(x-c)} = \frac{a^2 \ln(x-a)}{(a-b)(a-c)} + \frac{b^2 \ln(x-b)}{(b-c)(b-a)} + \frac{c^2 \ln(x-c)}{(c-a)(c-b)} + \mathcal{C}.$$

142.

$$\int \frac{(x-a)(x-b)}{(x-c)(x-d)} dx = \frac{1}{c-d} [(a-c)(b-c) \ln(x-c) - (a-d)(b-d) \ln(x-d)] + \mathcal{C}.$$

143.

$$\int \frac{(x^2+a^2)(x^2+b^2)}{(x^2+c^2)(x^2+d^2)} dx = \frac{1}{c^2-d^2} \left[\frac{(a^2-d^2)(b^2-d^2)}{d} \tan^{-1} \frac{x}{d} - \frac{(a^2-c^2)(b^2-c^2)}{c} \tan^{-1} \frac{x}{c} \right] + \mathcal{C}.$$

144.

$$64 \int \frac{\coth^4 x - 1}{\sinh x} dx = \text{sech}^4 \frac{x}{2} - \text{csch}^4 \frac{x}{2} - 10 \text{sech}^2 \frac{x}{2} - 10 \text{csch}^2 \frac{x}{2} - 40 \ln \left| \tanh \frac{x}{2} \right| + \mathcal{C}.$$

145.

$$8 \int \frac{dx}{x^4 - 2x^2 + 2} = \frac{1}{\sqrt{\sqrt{2}-1}} \tan^{-1} \frac{x^2 - \sqrt{2}}{x\sqrt{2(\sqrt{2}-1)}} - \frac{1}{\sqrt{\sqrt{2}-1}} \ln \left| \frac{x^2 - x\sqrt{2(\sqrt{2}-1)} + \sqrt{2}}{x^2 + x\sqrt{2(\sqrt{2}-1)} + \sqrt{2}} \right| + \mathcal{C}.$$

146.

$$\int \frac{\ln^3 x dx}{x^2 + 2x + 1} = 6 \text{Li}_3(-x) - 6 \text{Li}_2(-x) \ln x + \ln^2 x \left(\frac{x \ln x}{x+1} - \ln |\cos x| \right) + \mathcal{C}.$$

147.

$$\int \frac{1 + \tan x}{1 + \cot^3 x} dx = x + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{1 - 2 \tan x}{\sqrt{3}} \right) + \frac{1}{2} \ln |\sec x (\sec x - \sin x)| + \mathcal{C}.$$

148.

$$\int \frac{\sinh x dx}{1 + \text{sech } x + \text{sech}^2 x} = \cosh x + \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{\sqrt{3}}{2 \cosh x + 1} \right) - \frac{1}{2} \ln |\cosh 2x + 2 \cosh x + 3| + \mathcal{C}.$$

149.

$$\int \frac{x + \cos x}{1 + \sin x} dx = \ln |\sin x + 1| + 2 \ln \left| \sin \left(\frac{2x + \pi}{4} \right) \right| - x \cot \left(\frac{2x + \pi}{4} \right) + \mathcal{C}.$$

150.

$$\int \frac{\sec x dx}{\sin x + \cos^2 x} = \frac{1}{2} \ln \left| \frac{\sin^2 x - \sin x + 1}{\cos^2 x} \right| + \frac{1}{2\sqrt{5}} \ln \left| \frac{1 - \sqrt{5} - 2 \sin x}{1 + \sqrt{5} - 2 \sin x} \right| + \mathcal{C}.$$

§ Definite Integrals

151.

$$\int_0^\infty \frac{(1+2x)^{-1/3} - (1+x)^{-2/3}}{x^2} dx = 1 - \frac{2\pi}{3\sqrt{3}} + \frac{2\ln 2}{3}.$$

152.

$$\int_0^1 dx \int_x^1 dy \int_y^1 e^{z^3} dz = \frac{e-1}{6}.$$

153.

$$\int_0^{\frac{\pi}{4}} \frac{1}{\sin 2x + \cos 2x + 1} dx = \frac{\ln 2}{2}.$$

154.

$$\int_0^{\frac{\pi}{3}} (\sec x + \csc x) \left(\frac{\sec x + \tan x}{\csc x + \cot x} \right) dx = 1 + \frac{2}{\sqrt{3}}.$$

155.

$$\int_0^{\frac{1}{2}} \frac{x \sin^{-1}(x)}{\sqrt{1-x^2}} dx = \frac{6 - \sqrt{3}\pi}{12}.$$

156.

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{e^{\frac{1}{x}} + 1} dx = 1.$$

157.

$$\int_0^{\frac{\pi}{4}} \sqrt[3]{\tan x} dx = \frac{\pi\sqrt{3} - \ln 8}{6}.$$

158.

$$\int_0^\infty \frac{dx}{1 + \left(1 - \frac{1}{x}\right)^{2022}} dx = \frac{1}{2}.$$

159. Wallis Integral:

$$W_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{\Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{n}{2} + 1\right)} \sqrt{\pi}.$$

160.

$$\int_0^1 \frac{\ln x}{x-1} dx = \frac{\pi^2}{6}.$$

161.

$$\int_{-\infty}^\infty \frac{\sin x}{x} dx = \int_{-\infty}^\infty \frac{\sin^2 x}{x^2} dx = \pi.$$

162.

$$\int_0^{\frac{\pi}{2}} \ln(\cos x) dx = -\frac{\pi}{2} \ln 2.$$

163.

$$\int_0^1 \frac{\ln x}{x+1} dx = -\frac{\pi^2}{12}.$$

164.

$$\int_0^1 \sqrt[3]{2x^3 - 3x^2 - x + 1} dx = 0.$$

165.

$$\int_0^2 \frac{\ln(x+1)}{x^2 - x + 1} dx = \frac{\pi \ln 3}{2\sqrt{3}}.$$

166.

$$\int_0^1 \frac{\ln(1+x^2)}{1+x} = \frac{3 \ln^2(2)}{4} - \frac{\pi^2}{48}.$$

167.

$$\int_0^{\frac{\pi}{2}} \cos^{2022} x \cos(2022x) dx = \frac{\pi}{2^{2023}}.$$

168.

$$\int_0^1 \ln^{2022} x dx = 2022!.$$

169.

$$\int_1^e \frac{\ln x}{\sqrt{x}} dx = 4 - 2\sqrt{e}.$$

170.

$$\int_0^{\frac{\pi}{2}} \ln(2 + \tan^2 x) dx = \pi \ln(\sqrt{2} + 1).$$

171.

$$\int_0^{\frac{\pi}{2}} \sin^{2020} x \sin(2022x) dx = \frac{1}{2021}.$$

172.

$$\int_0^{\pi} \frac{1 - \sin x}{1 + \sin x} dx = 4 - \pi.$$

173.

$$\int_0^{\frac{\pi}{3}} \frac{\sin 2x}{2 + \cos x} dx = 1 + 4 \ln \frac{5}{6}.$$

174.

$$\int_0^{\infty} \frac{x - \sin x}{x^3(x^2 + 4)} dx = \frac{\pi}{32} (1 - e^{-2}).$$

175.

$$\int_0^{\frac{\pi}{2}} \frac{\cos x}{2 - \sin 2x} dx = \frac{\pi}{4}.$$

176.

$$\int_0^{\infty} \frac{x}{e^x + 1} dx = \int_0^{\infty} \ln(1 + e^{-x}) dx = \frac{\pi^2}{12}.$$

177.

$$\int_0^1 \frac{x-1}{(1+x^3) \ln x} dx = \frac{\ln 3}{2}.$$

178. Prove that

$$\int_0^{\infty} \prod_{k=1}^n \frac{\sin\left(\frac{x}{2^{k-1}}\right)}{\frac{x}{2^{k-1}}} dx = \frac{\pi}{2}$$

for $n = 1, 2, 3, 4, 5, 6$ & 7 .

179.

$$\int_{\sqrt{3}-1}^{2\sqrt{2}-1} \frac{1}{(x+1)\sqrt{x^2+2x+2}} dx = \ln \sqrt{\frac{3}{2}}.$$

180.

$$\int_0^{\frac{\pi}{2}} \frac{\ln(\tan x)}{1 - \tan x + \tan^2 x} dx = -\frac{7\pi^2}{72}.$$

181.

$$\int_1^e \frac{x^4 + 81}{x(x^2 + 9)^2} dx = \frac{1}{10} + \frac{9}{e^2 + 9}.$$

182.

$$\int_1^2 \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx = \frac{\ln 72}{10}.$$

183.

$$\int_0^\pi e^{-x} \sin^2(2x) dx = \frac{8 - 8e^{-\pi}}{17}.$$

184.

$$\int_1^\infty \frac{\ln^{627} x}{x^{2022}} dx = \frac{627!}{2021^{628}}.$$

185.

$$\int_0^{\frac{\pi}{6}} \frac{\sec^2 x}{\sqrt[2]{(\sec x + \tan x)^5}} dx = \frac{30 - 8\sqrt[4]{3}}{63}.$$

186.

$$\int_0^{\pi/6} \frac{\tan^4 \theta}{1 - \tan^2 \theta} d\theta = \frac{\pi}{12} - \frac{\sqrt{3}}{3} + \frac{\coth^{-1} \sqrt{3}}{2}.$$

187.

$$\int_0^1 \frac{1 - x^{99}}{(1+x)(1+x^{100})} dx = \frac{99}{100} \ln 2.$$

188.

$$\int_{-\infty}^{\infty} \frac{dx}{x^4 + x^3 + x^2 + x + 1} = \frac{\sqrt{10 + 2\sqrt{5}}\pi}{5}.$$

189.

$$\int_{-\infty}^{\infty} \int_t^{\infty} e^{\frac{1}{2}t^2 - x^2 + xt} dx dt = \int_{-\infty}^{\infty} \int_t^{\infty} e^{\frac{1}{2}x^2 - t^2 + xt} dx dt = \pi.$$

190.

$$\int_{-\infty}^{\infty} \operatorname{sech} x dx = \pi.$$

191.

$$\oint \frac{dz}{z} = 2\pi i.$$

192.

$$\int_0^\infty \ln \left(1 + \frac{1}{x^2} \right) = \pi.$$

193.

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2} = \frac{22}{7} - \pi.$$

194.

$$\int_0^1 \frac{x^4(1+x)^4}{1+x^2} = \pi - \frac{17}{15}.$$

195.

$$\int_0^{\infty} \frac{x^{\alpha-1}}{x+1} = \frac{\pi}{\sin \pi \alpha}, \quad 0 < \alpha < 1.$$

196.

$$\int_0^{\frac{\pi}{2}} x \ln(\sin x) dx = \frac{1}{16}(7\zeta(3) - \pi^2 \ln 4).$$

197.

$$\int_0^1 \ln x \sinh x dx = \gamma - \text{Chi}(1).$$

198.

$$\int_0^{\frac{\pi}{2}} \frac{\ln(\tan x)}{1 + \tan^3 x} dx = -\frac{37\pi^2}{432}.$$

199.

$$\int_0^{\infty} \frac{\sqrt{e^x - 1}}{2 \cosh x - 1} dx = \frac{\pi}{\sqrt{3}}.$$

200.

$$\int_0^1 \ln^3 \left(\frac{1-x}{1+x} \right) dx = -9\zeta(3).$$

201.

$$\int_0^{\infty} \sin(x^2) dx = \int_0^{\infty} \cos(x^2) dx = \sqrt{\frac{\pi}{8}}.$$

202.

$$\int_0^{\infty} \frac{\ln(2e^x - 1)}{e^x - 1} = \frac{\pi^2}{4}.$$

203.

$$\int_0^{\frac{\pi}{4}} x \left(\prod_{k=1}^{\infty} \cos \left(\frac{x}{2^k} \right) \right) dx = 1 - \frac{1}{\sqrt{2}}.$$

204.

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1} dx = \frac{\pi}{e}.$$

205.

$$\int_{-\infty}^{\infty} \frac{\sin(\tan x)}{x} dx = \pi \left(1 - \frac{1}{e} \right).$$

206.

$$\int_0^{\infty} \frac{\ln x \sin x}{x} dx = -\frac{\gamma \pi}{2}.$$

207.

$$\int_1^{\infty} \frac{\tan^{-1} \sqrt{2+x^2}}{(1+x^2)\sqrt{2+x^2}} dx = \frac{\pi^2}{32}.$$

208.

$$\int_0^{\infty} \frac{e^{\cos x} \cos(\sin x)}{1+x^2} dx = \pi e^{\frac{1}{e}}.$$

209.

$$\int_0^{\infty} \frac{\sin x \cos x}{\sinh x \cosh x} dx = \frac{\pi}{4} \tanh \frac{\pi}{2}.$$

210.

$$\int_0^{\infty} \ln^2 \left(\frac{1+x^2}{1+x^4} \right) dx = \frac{3\sqrt{2}\pi^3}{32}.$$

211.

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{\sin^{2k-1} x + \cos^{2k-1} x}{\sin^{2k} x + \cos^{2k} x} \right)^2 dx = \frac{2k-1}{2k^2} \frac{\pi}{\sin \frac{\pi}{2k}}.$$

212.

$$\int_0^1 \ln \left(\frac{1 - \sqrt{2}x + x^2}{1 - \sqrt{3}x + x^2} \right) \frac{1}{x} dx = \frac{19\pi^2}{288}.$$

213.

$$\int_0^{\frac{\pi}{2}} \frac{\sin x \ln(\sin x)}{\sqrt{1 + \sin^2 x}} dx = \frac{\pi}{8} \ln \frac{1}{2}.$$

214.

$$\int_0^\infty \frac{\sin(2 \ln^2 x)}{(1+x) \ln^2 x} dx = \sqrt{\pi}.$$

215.

$$\int_0^\infty \frac{\cos(\ln x)}{1+x+x^2} dx = 0.$$

216.

$$\int_0^1 \sqrt{\frac{x}{1-x}} \ln \left(\frac{x}{1-x} \right) dx = \pi.$$

217.

$$\int_0^{\frac{\pi}{2}} \frac{e^{\cos 2x} \sin(x + \sin 2x)}{\sin x} dx = \frac{\pi e}{2}.$$

218.

$$\int_0^1 \sin(\ln x) \frac{x^\alpha - x^\beta}{\ln x} dx = \arctan(1 + \beta) - \arctan(1 + \alpha).$$

219.

$$\int_1^\infty \frac{\ln(\ln x)}{x^2 + 2x \cos \alpha + 1} dx = \frac{\pi}{\sin \alpha} \ln \left| \frac{(2\pi)^{\frac{\alpha}{\pi}} \Gamma \left(\frac{1}{2} + \frac{\alpha}{2\pi} \right)}{\Gamma \left(\frac{1}{2} - \frac{\alpha}{2\pi} \right)} \right|.$$

220.

$$\int_0^1 \int_0^1 \ln \left(\frac{1}{e^x + e^y} \right) dx dy = \frac{3}{2} \zeta(3) + 2 \operatorname{Li}_3(-e) + \frac{\pi^2}{6} - \frac{1}{3}.$$

221.

$$\int_0^{\frac{\pi}{4}} \sqrt{\tan x (1 - \tan x)} dx = \pi \left(\sqrt{\frac{\sqrt{2} + 1}{2}} - 1 \right).$$

222.

$$\int_0^\infty \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}.$$

223.

$$\int_0^1 \frac{\ln x}{x^a - 1} dx = \frac{1}{a^2} \zeta \left(2, \frac{1}{a} \right).$$

224.

$$\int_0^1 \frac{\ln x}{x^a + 1} dx = \frac{1}{a^2} \left(\zeta \left(2, \frac{a+1}{2a} \right) - \zeta \left(2, \frac{1}{2a} \right) \right).$$

225.

$$\frac{\pi^3}{7} \int_0^1 \frac{x(1-x)}{\sin(\pi x)} dx = \int_0^1 \frac{\ln^2 x}{1-x} dx = \frac{8\pi^2}{7} \int_0^1 \frac{x(x^4 - 2x^2 + 1) \ln \ln \frac{1}{x}}{(1+x^4)^2} dx = \zeta(3).$$

226.

$$-\frac{1}{2} \int_0^1 \int_0^1 \frac{\ln(xy)}{1-xy} dx dy = \int_0^1 \int_0^1 \int_0^1 \frac{L_n(x)L_n(y)}{1-(1-xy)z} dx dy dz = \zeta(3).$$

227.

$$\int_0^\infty \sqrt{\operatorname{sech} x} dx = 4\sqrt{\frac{2}{\pi}} \Gamma\left(\frac{5}{4}\right)^2.$$

228.

$$\int_0^1 \frac{\ln(1+x^2)}{1+x^2} dx = \frac{1}{2}\pi \ln 2 - \mathcal{G}.$$

229.

$$\int_0^1 \frac{\arctan x \ln x}{x} dx = -\frac{\pi^3}{32}.$$

230.

$$\int_0^1 \frac{x \ln x}{1-\ln x} dx = -e^2 \operatorname{Ei}(-2) - \frac{1}{2}.$$

231.

$$\int_0^1 \int_0^1 \frac{x-1}{(1-xy)\ln(xy)} dx dy = \int_0^1 \frac{1}{1+x} \sum_{k=1}^\infty x^{2^k-1} dx = -\int_0^1 \ln \ln \frac{1}{x} dx = \gamma.$$

232.

$$\int_0^\infty \left(\frac{1}{1-e^{-x}} - \frac{1}{x} \right) e^{-x} dx = \int_0^\infty \frac{1}{x} \left(\frac{1}{1+x} - e^{-x} \right) dx = \gamma.$$

233.

$$\int_0^\infty \left(\frac{\tanh x}{x^3} - \frac{\operatorname{sech}^2 x}{x^2} \right) dx = \frac{7}{\pi^2} \zeta(3).$$

234.

$$\int_0^1 \operatorname{Li}_2\left(\frac{x}{x-1}\right) \frac{\ln x}{x} dx = \frac{\pi^4}{72}.$$

235.

$$\int_0^{\frac{\pi}{2}} \arctan\left(\frac{2a \sin^2 x}{1-a^2 \sin^2 x}\right) dx = \pi \arctan \frac{a}{2}.$$

236.

$$\int_0^\pi \frac{x^2}{\sqrt{5}-2\cos x} dx = 2\pi \ln^2 \phi + \frac{\pi^3}{15}.$$

237.

$$\int_0^{\frac{\pi}{2}} \frac{x \ln(\sin x)}{\sin 2x} dx = -\frac{\pi^3}{48}.$$

238.

$$\int_0^\infty \frac{\cos(2\pi x^2)}{\cosh^2(\pi x)} dx = \frac{1}{4}.$$

239.

$$\int_0^\infty \left(\prod_{k=1}^\infty \frac{k^2}{k^2+x^2} \right) dx = \frac{\pi}{2} \cdot \frac{n}{2n-1}.$$

240.

$$\int_0^\infty \frac{\cos(x^2) - \cos x}{x} dx = \frac{\gamma}{2}.$$

241.

$$\int_0^{\infty} \frac{\ln x dx}{(\pi^2 + \ln^2 x)(1+x)^2 \sqrt{x}} = -\frac{\pi}{24}.$$

242.

$$\int_0^{\infty} e^{-x} \ln x dx = -\gamma.$$

243.

$$\int_0^{\infty} e^{-x} \ln^2 x dx = \frac{\pi^2}{6} + \gamma^2.$$

244.

$$\int_0^{\infty} e^{-x} \ln^3 x dx = -2\zeta(3) - \frac{\pi^2 \gamma}{2} - \gamma^3.$$

245.

$$\int_0^{\infty} e^{-x} \ln^4 x dx = \frac{3\pi^4}{80} + 8\zeta(3) + \pi^2 \gamma^2 + \gamma^4.$$

246.

$$\int_{-\infty}^{\infty} \frac{1 - \cos x}{x^2} dx = \frac{4}{3} \int_{-\infty}^{\infty} \frac{\sin^3 x}{x^3} dx = \frac{3}{2} \int_{-\infty}^{\infty} \frac{\sin^4 x}{x^4} dx = \pi.$$

247.

$$\int_0^1 \frac{\ln x dx}{1 - x + x^2 - x^3} = \frac{\mathcal{G}}{2} - \frac{3}{32} \pi^2.$$

248.

$$\int_0^{\infty} \int_0^{\infty} \frac{\ln x \ln y}{\sqrt{xy}} \cos(x+y) dx dy = (\gamma + \ln 4) \pi^2.$$

249.

$$\int_0^1 \ln x \ln(1 - \sqrt[3]{x^2}) \sqrt[6]{1 - \sqrt[3]{x^2}} dx = \frac{1532}{375} - \frac{32}{25} \ln 2 - \frac{3}{10} \pi^2.$$

250.

$$\int_0^{\infty} \frac{\cos x}{x} \left(\int_0^x \frac{\sin t}{t} dt \right)^2 dx = -\frac{7}{6} \zeta(3).$$

251.

$$\int_0^{\frac{\pi}{2}} \frac{\sin(2x) \sec^4 x}{e^{2\pi \tan x} - 1} dx = \frac{1}{12}.$$

252.

$$\int_0^1 \frac{(\arctan x)^2}{x} dx = \frac{\pi \mathcal{G}}{2} - \frac{7}{8} \zeta(3).$$

253.

$$8 \int_0^1 \int_0^1 \frac{xy \ln(1-xy)}{(1+xy)^3} dx dy = 2 \ln^2 2 + 4 \ln 2 - \frac{5\pi^2}{12}.$$

254.

$$\int_0^{\infty} \frac{\ln x dx}{\cosh^2 x} = \ln \frac{\pi}{4} - \gamma.$$

255.

$$8 \int_0^{\frac{\pi}{4}} \arctan(\cot^2 x) dx = \pi^2 - 2 \ln^2(\sqrt{2} + 1).$$

256.

$$\int_0^{\infty} \frac{x}{x^8 + 2x^4 + 1} = -2 \int_0^{\infty} \frac{x \ln x}{x^8 + 2x^4 + 1} = 2 \int_0^{\infty} \frac{x^5 \ln x}{x^8 + 2x^4 + 1} = \frac{\pi}{8}.$$

257.

$$\int_0^{\frac{\pi}{4}} \int_0^{\frac{\pi}{4}} \arctan(\cos x \cot y) \sec^3 x \tan^2 y \sec^2 y dx dy = \frac{1}{3}.$$

258.

$$\int_0^\infty \frac{\ln^2 x}{x^2 + 2x + 2} dx = \frac{5\pi^3 + 4\pi \ln^2 2}{64}.$$

259.

$$\int_0^\infty \frac{\ln(1+x^2)}{(1+x^2)^3} dx = \frac{\pi(12\ln 2 - 7)}{32}.$$

260.

$$\int_0^{\frac{\pi}{2}} \frac{\arctan(2\cos^2 x)}{\cos^2 x} dx = \sqrt{\frac{\sqrt{5}-1}{2}} \pi.$$

261.

$$\int_0^1 \frac{x^3 \ln x}{(1+x^2)^2} dx = \frac{\ln 2}{4} - \frac{\pi^2}{48}.$$

262.

$$\int_0^1 \int_0^1 \int_0^1 \frac{dx dy dz}{2 - xyz} = \frac{7}{8} \zeta(3) + \frac{1}{6} \ln^3 2 - \frac{\pi^2}{12} \ln 2.$$

263.

$$\int_0^\infty \operatorname{sech} x \ln x dx = \frac{\pi}{2} \ln \left(\frac{\Gamma(\frac{3}{4})^4}{\pi} \right).$$

264.

$$\int_0^1 \int_0^1 \sqrt{x^2 + y^2} dx dy = \frac{\sqrt{2}}{3} + \frac{1}{3} \ln(1 + \sqrt{2}).$$

265.

$$\int_0^\infty \frac{x^{s-1}}{(x^b + \sqrt{x^{2b} + 1})^a} = \frac{a}{b} \frac{\Gamma(\frac{s}{b}) \Gamma(\frac{a}{2} - \frac{s}{2b})}{2^{\frac{s}{b}+1} \Gamma(\frac{a}{2} + \frac{s}{2b} + 1)}.$$

266.

$$\int_0^\infty e^{-x^2} \ln x dx = -\frac{\sqrt{\pi}}{4} (\gamma + 2 \ln 2).$$

267.

$$\int_0^\infty \arctan\left(\frac{t}{x}\right) \sin x dx = \frac{\pi}{2} - \frac{\pi}{2} e^{-t}.$$

268.

$$\int_0^\infty \frac{1+x^4 \ln x}{1+x^6} dx = \frac{\pi}{3} + \frac{\sqrt{3}}{18} \pi^2.$$

269.

$$\int_0^\infty \frac{\cos(ax)}{x^2 + b^2} dx = \frac{\pi}{2b} e^{-ab}.$$

270.

$$\int_0^\infty \frac{\ln x dx}{x^4 + 2x^2 + 4} = \frac{\pi \ln 2}{12\sqrt{3}}.$$

271.

$$\int_0^\infty \frac{1}{\sqrt{x}} \sin\left(\pi^2 x + \frac{1}{x}\right) dx = \frac{1}{\sqrt{2\pi}}.$$

272. Prove that

$$\int_0^{\sin^2 x} \arcsin \sqrt{t} dt + \int_0^{\cos^2 x} \arccos \sqrt{t} dt = \frac{\pi}{4}.$$

273.

$$\int_0^1 \int_0^1 \int_0^1 \ln(x + xy + xyz) dx dy dz = \ln 27 - \ln 8 - 3 - \text{Li}_2(-2) - \frac{\pi^2}{12}.$$

274.

$$\int_{-\infty}^{\infty} \frac{(x^3 - 4x) \sin x + (3x^2 - 4) \cos x}{(x^3 - 4x)^2 + \cos^2 x} dx = -3\pi.$$

275.

$$\int_0^{\infty} \frac{x e^{-2x}}{e^{-x} + 1} dx = 1 - \frac{\pi^2}{12}.$$

276.

$$\int_0^{\frac{\pi}{2}} \sin(\cot^2 x) \sec^2 x dx = \sqrt{\frac{\pi}{2}}.$$

277.

$$\int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\tan x}}{(\sin x + \cos x)^2} dx = \frac{2\sqrt{3}\pi}{9}.$$

278.

$$\int_0^{\infty} \sin x \text{Si}(x) dx = \int_0^{\infty} \cos x \text{Ci}(x) dx = -\frac{\pi}{4}.$$

279. The function defined by the equality

$$J_0(x) = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \cos(x \sin \theta) d\theta$$

is called *Bessel's function* of zeroth order. Prove that

$$\int_0^{\infty} e^{-ax} J_0(x) dx = \frac{1}{\sqrt{1+a^2}}, \quad a > 0.$$

$$J_0''(x) + \frac{J_0'(x)}{x} + J_0(x) = 0.$$

280. Prove that the function

$$y = \int_{-1}^1 (z^2 - 1)^{n-1} e^{xz} dz$$

satisfies the differential equation

$$xy' + 2ny' - xy = 0.$$

281. Prove that the function

$$y = \int_0^{\infty} \frac{e^{-xz} dz}{(1+z^2)^{n+1}}$$

satisfies the differential equation

$$xy'' - 2ny' + xy = 1.$$

282.

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{(\pi - 4x) \tan x}{1 - \tan x} dx = \pi \ln 2 - \frac{\pi^2}{4}.$$

283.

$$\int_{-\infty}^{\infty} \frac{1}{\cosh x} \frac{dx}{(e^x - x)^2 + \pi^2} = \frac{4}{2 + \pi} - \frac{2\mathcal{W}(1)}{(1 + \mathcal{W}(1))(1 + \mathcal{W}^2(1))}.$$

284.

$$\int_0^1 \int_0^1 \int_0^1 \frac{x^4 y^3 z^2}{(x+y+z)(x^2+y^2+z^2)-(x^3+y^3+z^3)} dx dy dz = \frac{1}{144}$$

285.

$$\int_0^\infty \frac{\ln x \cdot (1 - \cos x)}{x^2} dx = \frac{\pi}{2}(1 - \gamma).$$

286.

$$\int_0^{\frac{\pi}{2}} \frac{x(\pi - x)}{\sin x} dx = \frac{7}{2}\zeta(3).$$

287. Let $n \in \mathbb{N}$ & $\theta \in (0, \frac{\pi}{2})$. Prove that

$$\int_0^\pi \sin(n\alpha) \arctan\left(\frac{\tan(\frac{\alpha}{2})}{\tan(\frac{\theta}{2})}\right) d\alpha = \frac{\pi}{2n} \tan^n\left(\frac{\pi}{4} - \frac{\theta}{2}\right) - \frac{\pi(-1)^n}{2n}.$$

288.

$$\int_0^1 \ln x \ln\left(\frac{1}{x^2} + 1\right) dx = -2\mathcal{G} - \frac{\pi}{2} - \ln 2.$$

289.

$$\int_0^1 \frac{\zeta(m) - \text{Li}_m(x)}{1-x} (\ln x)^{m-1} dx = (-1)^{m-1} \zeta^2(m) \Gamma(m) + (-1)^m (m-1)! \sum_{n=1}^\infty \left(\frac{\mathcal{H}_n}{n+1}\right)^m.$$

290. Let $\mathbb{R} \ni s \geq 2$. Prove that

$$\int_0^\infty \frac{x^{\frac{s}{2}-1}}{1+x^s} \text{Li}_2(-x) dx = -\frac{\pi^3}{4} \left(\frac{1}{3s} + \frac{1}{s^3}\right).$$

291.

$$\int_0^1 \frac{\ln x \ln(1-x) \ln^2(1+x)}{x} dx = \frac{7\pi^2}{48} \zeta(3) - \frac{25}{16} \zeta(5).$$

292.

$$\int_0^1 \ln^2 x \arctan x dx = \frac{\pi}{2} - \frac{\pi^2}{24} - \ln 2 - \frac{9\zeta(3)}{48}.$$

293.

$$\int_0^1 \frac{\ln x \ln(1+x)}{x} dx = -\frac{3\zeta(3)}{4}.$$

294.

$$\int_0^\infty x(\ln x)^3 e^{-x} dx = (1-\gamma)^3 + 3(1-\gamma) \left(\frac{\pi^2}{6} - 1\right) - 2\zeta(3) + 2.$$

295.

$$\int_{-\infty}^\infty x^2 \operatorname{sech}\left(\frac{x^4 - 10x^2 + 6x + 7}{x^3 - 7x + 6}\right) dx = \frac{\pi^3}{4} + 3\pi.$$

296.

$$\int_0^\infty \frac{x \sin x + \cos x - 1}{x^2} e^{-x^2} dx = \sqrt{\pi} \left(1 - \frac{1}{\sqrt[4]{e}}\right).$$

297.

$$\int_0^1 \int_0^{\frac{\pi}{2}} \frac{x^3(1+\cos^2 y)}{\sqrt{4-x^4(1+\cos^2 y)^2}} dy dx = \frac{\arctan \sqrt{2}}{\sqrt{2}} - \frac{\ln 3}{4}$$

298. Let α & $\beta \geq 1$. Prove the following identity

$$\int_0^1 \frac{(1-t_0)^{\beta-1}}{t_0^\alpha} dt_0 \int_0^{t_0} \frac{dt_1}{t_1} \cdots \int_0^{t_{n-2}} \frac{dt_{n-1}}{t_{n-1}} \int_0^{t_{n-1}} \frac{t_n^{\alpha-1}}{(1-t_n)^\beta} dt_n = \sum_{l=0}^\infty \frac{1}{(l+\alpha)^n (l+\beta)}.$$

299.

$$\int_0^1 \frac{\text{Li}(x) \ln(-\ln x)}{x} dx = 1 + \gamma.$$

300.

$$\int_0^1 \frac{\ln(1+\alpha x)}{\sqrt{(1+\alpha x)^3}} \cdot \frac{dx}{\sqrt{1-x}} = -\frac{4}{\alpha(\alpha+1)} \left(\alpha - \sqrt{\alpha} \arccos \left(\frac{1}{\sqrt{\alpha+1}} \right) \right).$$

§ Important Symbols & Notations Used

- Li - Polylogarithmic Function/Polylogarithm

A Polylogarithm of order m is a complex-valued function of a variable s defined as

$$\text{Li}_m(s) = \sum_{k=1}^{\infty} \frac{s^k}{k^m}.$$

- Ei - Exponential Integral

An Exponential Integral of a non-zero variable x is defined over the complex plane as

$$\text{Ei}(x) = - \int_{-x}^{\infty} \frac{e^{-t}}{t} dt = \int_{-\infty}^x \frac{e^t}{t} dt.$$

- erf - Error Function/Gauss Error Function

An Error Function of variable z is defined over the Complex Plane as

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt.$$

- $\mathcal{L}$ - Laplace Transform

The Laplace Transform of a function $f(t)$ with respect to variable s is defined as

$$\mathcal{L}_t(f(t))(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

- Ci - Cosine Integral

A Cosine Integral of a variable x is defined as

$$\text{Ci}(x) = - \int_{-\infty}^x \frac{\cos t}{t} dt.$$

- Si - Sine Integral

A Sine Integral of a variable x is defined as

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt.$$

- φ - Standard Normal Probability Density Function

It is defined as

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

- Φ - Cumulative Distribution Function

It is defined as

$$\Phi(x) = \int_{-\infty}^x \varphi(t) dt = \frac{1}{2} \left(1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right).$$

- Γ - Gamma Function

A Gamma Function is defined for all complex numbers z except non-positive integers.

$$\Gamma(n) = (n-1)!, \quad n \in \mathbb{N}; \quad \Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt, \quad \Re(z) > 0.$$

- $\oint$ - Closed Contour Integral/Line Integral

The closed contour integral is used to integrate a function along a closed path (or loop) in the complex plane or in a vector field. The $\oint$ symbol indicates that the path of integration forms a closed loop. This can be important for applications in physics and engineering, especially in fields like electromagnetism. Commonly used in complex analysis (for example, in Cauchy's integral theorem) and in calculating the circulation of a vector field around a closed curve.

- ζ - Riemann Zeta Function/Hurwitz Zeta Function

The Hurwitz Zeta Function is a complex-valued defined for a variable s and a where a isn't a non-positive integer as

$$\zeta(s, a) = \sum_{k=0}^{\infty} \frac{1}{(k+a)^s}.$$

For $a = 1$, it becomes the Riemann Zeta Function.

$$\zeta(s, 1) = \zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s}.$$

- Chi - Hyperbolic Cosine Integral

It is defined for a variable x as

$$\operatorname{Chi}(x) = - \int_x^{\infty} \frac{\cosh t}{t} dt.$$

- Shi - Hyperbolic Sine Integral

It is defined for a variable x as

$$\operatorname{Shi}(x) = \int_0^x \frac{\sinh t}{t} dt.$$

- $\mathcal{H}_n$ - n-th Harmonic Number

The n-th harmonic number is defined as

$$\mathcal{H}_n = \sum_{k=1}^n \frac{1}{k}.$$

- γ - Euler-Mascheroni Constant

The Euler-Mascheroni Constant is defined as the following limit

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \ln n \right) = \lim_{n \rightarrow \infty} (\mathcal{H}_n - \ln n).$$

- $\mathcal{G}$ - Catalan's Constant

The Catalan's Constant is defined as

$$\mathcal{G} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)^2}.$$

- ϕ - Golden Ratio

The Golden Ratio is defined the as a ratio whose value is unity more than it's reciprocal. It's the positive root of $x^2 - x - 1 = 0$, i.e.

$$\phi = \frac{\sqrt{5} + 1}{2}.$$

- $\mathcal{W}$ - Lambert W Function/Omega Function/Product Logarithm

The Lambert W Function is a multivaused function which is defined as the inverse function or the converse relation of

$$f(W) = We^W,$$

where W is a complex numbers,

- L_n - Legendre Polynomials/Legendre Functions of the First Kind

They are defined as the following closed contour integral over the Complex Plane

$$L_n(z) = \frac{1}{2\pi i} \oint \frac{dt}{t^{n+1} \sqrt{1-2t+t^2}}.$$

Thank You All!